

N A Adarsh Pritam

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 [Google Scholar](https://scholar.google.com/citations?user=...)

OVERVIEW

I am currently a **Research Intern at the Centre for Brain Research (CBR), Indian Institute of Science (IISc)**, where I primarily work on **machine learning methodologies for dementia research** (for early detection of cognitive decline). My research interests includes **Representation Learning, Generative Models, Computer Vision, Medical Imaging, Multimodal Learning, NeuroAI and Medical AI**.

RESEARCH EXPERIENCE

J: Journal C: Conference W: Workshop P: Preprint

CBRAIN Research Intern

May 2026 – Present

Indian Institute of Science (IISc) - Centre for Brain Research (CBR)

- Conducting research on machine and statistical learning methods for early prediction of Mild Cognitive Impairment (MCI) using multimodal longitudinal data, with focus on clinically interpretable risk modelling for Indian urban and rural population.
- Mentored clinicians and junior researchers in statistical analysis and machine learning methodologies to improve interdisciplinary collaboration between clinical and AI research teams.
- **Principle Investigator:** Dr. Thomas Gregor Issac

Graduate Researcher

Aug 2025 – May 2026

Alliance University

- Conducted research on how cortical representations in the brain align with vision-language latent spaces for brain-to-image reconstruction.
- Conducted research on generative models in medical imaging, designing a systematic benchmarking framework to evaluate Generative Adversarial Networks (GANs) and diffusion models for synthetic dermoscopic image augmentation and downstream melanoma diagnosis.
- **Supervisor:** Dr. Jeba Shiney O
- **Publications:** [P1](#) , [P2](#) , [C1](#)

ML Research Intern

May 2025 – Sep 2025

Indian Institute of Information Technology, Kottayam - Intelligent Integrated Circuits and Systems (i2CS) Research Group

- Developed a vision-language model for plant disease understanding to provide descriptive analysis from images of affected leaves.
- **Supervisor:** Dr. kala S

PUBLICATIONS

J: Journal C: Conference W: Workshop P: Preprint

P1 [Brain2VLM: Hierarchical Alignment Between Cortical Representations and Vision-Language Latent Spaces](#)

N A Adarsh Pritam, Jeba Shiney O, Sanyam Jain.
bioRxiv Preprint, 2026 (Currently under review).

P2 [SkinGenBench: Generative Model and Preprocessing Effects for Synthetic Dermoscopic Augmentation in Melanoma Diagnosis](#)

N A Adarsh Pritam, Jeba Shiney O, Sanyam Jain.
arXiv Preprint, 2025 (Currently under review).

C1 [American Sign Language to Fluent English Detection and Translation Using T5](#)

N A Adarsh Pritam, Asha Kurian.
IEEE International Conference on Emerging Technologies in Computing and Communication (IEEE - ETCC), 2025.

OPEN SOURCE PROJECTS

PyTorch Image Classification Framework

- Extended an open-source PyTorch image classification framework by integrating Grad-CAM visualization and support for modern convolutional and transformer architectures.
- **GitHub:** github.com/adarsh-crafts/pytorch-classification-extended

LLaMA from Scratch

- Implemented a LLaMA-style large language model from scratch in PyTorch, reproducing core Transformer components including grouped-query attention, rotary positional embeddings, and RMS normalization.
- **GitHub:** github.com/adarsh-crafts/llama-llm-from-scratch

EDUCATION

Alliance University, Bangalore

2024 – 2026

Master of Science in Data Science

CGPA: 8.6

Bengaluru City University, Bangalore

2024 – 2026

Bachelor of Science in Mathematics and Statistics

CGPA: 6.5

Relevant Coursework: Machine Learning I & II, Deep Learning, ML Techniques for Image Processing, Mathematical Foundations for Data Science, Natural Language Processing, Inferential and Descriptive Statistics, Foundations of Data Science, Applied Mathematics, Probability, Operations Research, ML-Operations (MLOps), Research Methodology

SKILLS

Programming: Python, R, SQL

Research Interests: Representation Learning, Generative Models, Computer Vision, NeuroAI, Medical AI, Deep Learning, Machine Learning, Statistical Learning

Core Libraries: PyTorch, Torchvision, OpenCV, Transformers, Scikit-learn, Optuna, Shap, Pandas, Numpy

Tools: Git, Docker, MLflow, ZenML, AWS, Weights&Biases, Latex

CERTIFICATIONS

- AIF-C01: AWS Certified AI Practitioner - Amazon Web Services (2024)
- Fundamentals of Deep Learning - NVIDIA (2025)
- Mathematics for Machine Learning and Data Science – DeepLearning.AI (2024)

ACHIEVEMENTS

- Invited to serve on the Board of Studies (BoS) curriculum committee for the MSc Data Science program, contributing to curriculum design for subsequent (2026 onward) student cohorts.
- Invited to deliver seminars on machine learning research methodology for MSc Data Science students.

REFERENCES

1. Dr. Thomas Gregor Issac (thomasgregor@cbr-iisc.ac.in), Associate Professor, Centre for Brain Research, Indian Institute of Science.
2. Dr. Jeba Shiney (jeba.shiney@alliance.edu.in), Professor, Alliance School of Advanced Computing, Alliance University.
3. Dr. Kala S (kala@iiitkottayam.ac.in), Assistant Professor & Head of i2CS Research Group, Indian Institute of Information Technology (IIIT) Kottayam.